

Spatial Augmented Reality Assistance System with Accelerometer and Projection Mapping at Cleaning Activities

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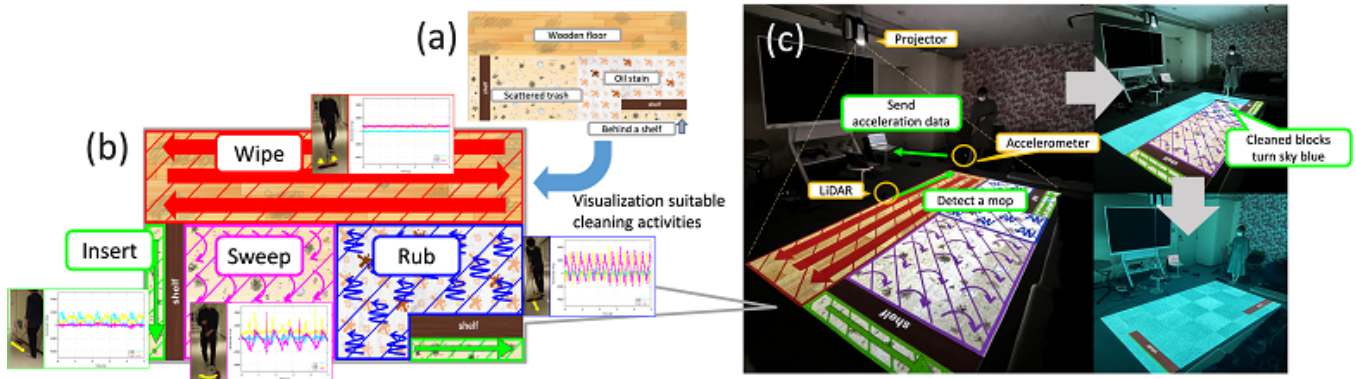


Figure 1: (a)Projected situation. (b)Visualized requirements. (c)Experimental setup and system overview.

CCS CONCEPTS

• **Human-centered computing** → Mixed / augmented reality; Mobile computing; • **Applied computing** → Computer-assisted instruction.

KEYWORDS

augmented reality, projection mapping, activity recognition, cyber-physical system

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1 INTRODUCTION

Projection mapping is a technological process where digital images are projected onto a physical surface using a video projector [Raskar et al. 2001]. It is one of the promising solutions of Spatial Augmented Reality (SAR) for well-being. Projection-based work supporting systems were investigated on an assembly sequence [Rodriguez et al.

2015] and hardware-related appropriation [Ludwig et al. 2019]. On-desktop projection have been proposed for learning support systems for math development [Beşevli et al. 2019] and typing on a physical keyboard [Sono and Hasegawa 2019]. On-body projection systems were developed for education [Hoang et al. 2017][Ferdous et al. 2019]. Although projection-based human-machine interfaces have attracted attention, there have been very few works on integration with sensor-based activity recognition.

Therefore, in this paper we propose an SAR assistance system with accelerometer and projection mapping. We considered cleaning activities [Yaegashi et al. 2022] for recognition to achieve well-being during and after the pandemic. The contribution of the proposed idea is to integrate SAR and real-time activity recognition technologies. The sensor-based activity recognition is advantageous in terms of privacy compared with well-known gesture recognition systems using a camera. The fixed floor-projection also contributes for disuse of a camera by avoiding dynamic calibration. A small and light-weight sensor device mounted on a cleaning tool does not affect the action. Low-latency feedback is achieved by an edge computing based architecture.

2 APPROACH

This paper proposed an SAR assistance system for cleaning activities. The goal of the proposed system is to provide visual assistance to a cleaning user based on real-time activity recognition technologies. The system configuration is depicted in Fig. 2. Projection mapping is employed to suggest expected actions to a user, i.e. predefined requirements are projected on the floors and walls in the room.

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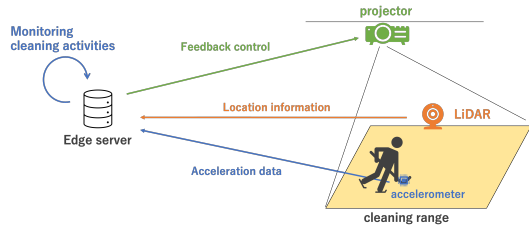


Figure 2: System configuration of proposed assistance system

The movement of a cleaning tool is tracked with LiDAR to identify the place of action. A tool-mounted sensor device collects 3-axis accelerometer data and send them to an edge server. The edge server recognizes the type of cleaning activity using a deep neural network such as Long Short Term Memory (LSTM). The projected images are updated in real-time based on the recognition results to imply the cleaning states, i.e. requirement satisfaction/dissatisfaction.

3 IMPLEMENTATION

A TWELITE CUE equipped with a 3-axis acceleration sensor MEMSIC MC3630 was attached to the shaft of a mop. The sampling rate was set to 100 Hz. The acquired accelerometer data were sent to the edge server using the TWELITE protocol based on IEEE802.15.4. The edge server recognized the actions using LSTM with accelerometer data during 2 sec time-window. We employed the inclusive list of cleaning activities using a mop [Yaegashi et al. 2022] that includes

- (1) *Wipe*: Wiping by pushing the mop.
- (2) *Sweep*: Sweeping like a broom.
- (3) *Rub*: Rubbing a certain floor area.
- (4) *Insert*: Inserting in a narrow space.
- (5) *Other*: Non-cleaning actions such as walking with a mop.

The inference results are sent to the other edge server via OpenSound Control (OSC) protocol every 0.5 sec. UST-20lx LiDAR from Hokuyo was installed to measure the current location of the mop. TouchDesigner installed on the edge server converted the detected location from polar to rectilinear coordinates.

We simulated a living environment in laboratory by projecting simulation image on a floor. The projected simulation image consisted of 4 situations as shown in Fig. 1; wooden floor, space behind the shelf, floors with oil stain, and scattered trash. The suitable actions for each situation were visually suggested to a user using WX4241N, a short focus video projector from RICOH installed near the ceiling. The target area was divided into 10×20 blocks to imply the cleaning states in real-time. The user's action was monitored with the received recognition results and coordinate information. If the preset activity requirement is satisfied in a block, TouchDesigner turned the block sky blue to visualize it.

4 EXPERIMENTAL RESULTS

The usability of the proposed system was evaluated in terms of effectiveness and satisfaction. Seven male and female participants performed cleaning with and without the proposed assistance. The

effectiveness was evaluated with the achievement rate defined as

$$\gamma = \frac{n_S}{n_A}, \quad (1)$$

where n_S denotes the number of blocks which satisfied requirement and n_A denotes the number of all blocks. The average γ without the proposed scheme was 81.3%. The proposed system improved γ to 93.4% by visualizing the current state. These results confirmed the effectiveness of the proposed idea. After cleaning, the participants were administered a five-grade evaluation questionnaire. The motivation enhancements without and with the assistance were approximately 1.6 and 4.7, respectively. We confirmed that SAR assistance improves user satisfaction and motivates users for cleaning activities.

5 CONCLUSION

The proposed idea leverages LiDAR and accelerometer data for real-time recognition of the place and types of cleaning activities. The contribution of this work is to integrate SAR and real-time activity recognition technologies for well-being during and after the pandemic. It was confirmed from the evaluation results using the implemented system that the recognition-based real-time visualization well supports the users. The sensor-based activity recognition has the merit of privacy compared with gesture recognition using a camera. It will be future work to collect a large-scale dataset and to develop coaching technologies for enabling users to perform more effective cleaning activities.

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